

Mod 4, Task 1_Case Analysis Form

CASE 1: Gender Shades. Section 1: Bias identified and how measured (include specific

Buolamwini and Gebru identified intersectional accuracy disparities in commercial gender-classification APIs. Using the Pilot Parliaments Benchmark labeled by perceived gender and Fitzpatrick skin type, they found all systems performed worst on darker-skinned female faces; error rates for darker females ranged from 20.8% to 34.7%, compared with 0.0% to 0.8% for lighter-skinned male faces.

Section 2 — Who was harmed and through what institutional mechanism:

Darker-skinned women and female-presenting people were most harmed. The immediate mechanism was misclassification or lower-confidence classification by facial-analysis systems, which could lead to exclusion, misidentification, or unequal treatment when these tools are embedded in higher-stakes institutional settings such as surveillance, hiring pipelines, authentication, or law enforcement.

Section 3 — Root cause classification (check applicable + one sentence evidence per category):

- ☒ Representation bias — Existing benchmark datasets were skewed toward lighter-skinned and male faces; darker females were underrepresented, so systems could appear accurate overall while failing for this subgroup.
- ☒ Measurement bias — Skin tone was approximated from images using Fitzpatrick categories and the systems reduced gender to binary labels, limiting how accurately key attributes were measured.
- ☒ Aggregation bias — Overall accuracy masked subgroup error; the same model-performance summary hid much higher error for darker females.
- ☒ Deployment bias — Although the paper audits APIs, the harm becomes consequential when face-

Section 4 — Remediation proposed/implemented — and was it sufficient? Why or why not?

Remediation proposed: more balanced benchmark datasets, intersectional subgroup reporting, transparency about training and benchmark composition, and accountability for closing performance gaps. This is stronger than a narrow patch because it targets representation and evaluation practices, but it is not fully sufficient unless vendors also change model development, disclose subgroup metrics,

CASE 2: Obermeyer et al. (2019) — racial bias in healthcare cost-prediction algorithms. Section 1 — Bias identified and how measured:

Obermeyer et al. identified racial bias in a widely used health care risk-prediction algorithm. At the same predicted risk score, Black patients were sicker than White patients; at the 97th-percentile threshold, Black patients had 26.3% more chronic illnesses (4.8 vs. 3.8). Correcting the disparity would increase the share of Black patients receiving additional care-management support from 17.7% to 46.5%.

Section 2 — Who was harmed and through what institutional mechanism:

Black patients were most harmed. The institutional mechanism was under-referral to high-risk care management: because the model used predicted health care cost as a proxy for health need, it underestimated the needs of Black patients who historically generated lower health care expenditures at similar illness levels, resulting in fewer extra resources, clinician attention, and care coordination.

Section 3 — Root cause classification:

Historical bias: Unequal access to care and structural racism meant historical cost data reflected differences in care received, not just medical need.

Measurement bias: The label measured future cost rather than illness or need, so the target variable systematically underestimated Black patients' health burden.

Aggregation bias: A single cost-risk model treated the relationship between cost and health as if it were equivalent across racial groups, even though Black patients were sicker at the same score.

Deployment bias: The biased score was used to screen patients for scarce care-management resources, so a biased prediction translated into unequal institutional support.

Section 4 — Remediation proposed/implemented — sufficiency evaluation:

Remediation was proposed and partly implemented with the manufacturer by changing the label from future cost toward an index combining health prediction with cost prediction. That approach reduced excess chronic illness among Black patients at a given score by 84%, so it addresses a root cause—the flawed proxy label—rather than only the symptom. It is still incomplete without ongoing audits, monitoring after deployment, and governance over how risk scores allocate scarce care resources.

100-word comparative synthesis — shared root causes and differing harm mechanisms:

The shared root cause is not intentional discrimination but biased problem formulation and evaluation. Gender Shades shows representation and measurement bias: benchmark data and binary labels made aggregate accuracy look acceptable while darker-skinned women were misclassified. Obermeyer shows historical and measurement bias: health care cost was treated as a proxy for health need even though cost reflected unequal access to care. In both cases, aggregate performance hid subgroup harm. The harm mechanism differs: Gender Shades produces direct misclassification in facial-analysis systems, while Obermeyer produces unequal allocation of care-management resources through a